Kaustubh Dua

Software Developer · B.Tech. Electronics & Computer Engineering, JIIT Noida · Class of 2029

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PROFILE

Engineering student who ships full-stack products end to end: React or Next.js front ends, FastAPI and Supabase back ends, automated tests and CI. Careful about security, data boundaries and honest status labels. Open to software engineering internships.

EDUCATION

Jaypee Institute of Information Technology (JIIT), Noida
B.Tech., Electronics & Computer Engineering (ECM) · Expected graduation 2029

SKILLS

Languages Python · TypeScript · JavaScript · SQL · C++17
Build React · Next.js · FastAPI · Supabase · PostgreSQL / PostGIS · Google OR-Tools · scikit-learn
Ship Git · GitHub Actions · pytest · Playwright · Docker · CodeQL

PROJECTS

NIRDHOOM Crop-residue coordination platform · prototype
React 19, Supabase, PostGIS, OR-Tools, PWA

- Built the full workflow from a farmer's booking to operator jobs, dispatch and a buyer-facing pool of **verified residue lots**.
- Kept service keys **server-side** behind Row Level Security; added SHA-256 evidence hashing and an offline upload queue.

github.com/kaustubhdua/nihdhoom · Live: nihdhoom-9qiu.vercel.app

Saathi Voice-first Hindi and Hinglish assistant · MVP
Next.js, FastAPI, speech APIs, GitHub Actions

- Made intent routing **deterministic** and tested it on a **120-case benchmark**, so the language model is never the source of truth.
- Attached a source and timestamp to every factual answer; added consent prompts and **human escalation**.

github.com/kaustubhdua/Saathi · Live: saathi-two-pi.vercel.app

PatchPilot Pull-request risk triage · early stage
Python, FastAPI, pytest, Docker

- Wrote an analyser that turns a diff into **evidence-backed risk signals** and a 0-100 score, without ever running the code it reads.
- Hardened the API: input limits, strict schemas, rate limiting and a non-root Docker image.

github.com/kaustubhdua/PatchPilot

JIIT Youth Club website Public site for a campus club · live
React 19, Supabase, Playwright, CodeQL

- Built the site for events, clubs, gallery and recruitment, guarded by **browser tests and security scanning** on every change.

github.com/kaustubhdua/JYC-Website · Live: jyc-website-eight.vercel.app

TEAMWORK

Adapt AI (contributor) Local AI assistant that recommends an Ollama model for a machine's RAM and VRAM. Team project; my fork.